

The Riemann hypothesis is true up to $3 \cdot 10^{12}$

Dave Platt*

School of Mathematics,
University of Bristol, Bristol, UK
dave.platt@bris.ac.uk

Tim Trudgian*[†]

School of Science
The University of New South Wales Canberra, Australia
t.trudgian@adfa.edu.au

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Abstract

We verify numerically, in a rigorous way using interval arithmetic, that the Riemann hypothesis is true up to height $3 \cdot 10^{12}$. That is, all zeroes $\beta + i\gamma$ of the Riemann zeta-function with $0 < \gamma \leq 3 \cdot 10^{12}$ have $\beta = 1/2$.

1 Introduction

The Riemann zeta-function $\zeta(s)$ has trivial zeroes at $s = -2, -4, -6 \dots$, and non-trivial zeroes in the strip $0 < \sigma < 1$, where here, and hereafter $s = \sigma + it$. The Riemann hypothesis asserts that all non-trivial zeroes $\rho = \beta + i\gamma$ have $\beta = 1/2$. In the absence of a proof, it is extremely important to obtain partial verifications of the Riemann hypothesis. To that end, define H as the largest number for which it is known that all zeroes $\rho = \beta + i\gamma$ with $0 < \gamma \leq H$ have $\beta = 1/2$. This problem has a long history; for a glimpse of this we refer the reader to [11, p. 2].

In recent years, three calculations have been referenced frequently in the literature. The first result is by Wedeniwski¹ [36] in 2004, with $H = 2.41 \dots \cdot 10^{11}$. The second is by Gourdon

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¹It is difficult to pinpoint the height claimed in these calculations. The third slide of [36], dated 20th May 2003, claims the first 200 billion zeroes were checked, equating to $H = 5.72 \dots \cdot 10^{10}$. The end of [36] contains the statement that the project computed 385 billion zeroes or $H = 1.07 \dots \cdot 10^{11}$ if we assume these

[11], also in 2004, which establishes $H = 2.44 \dots \cdot 10^{12}$. The third is by the first author in 2017 [21] (see also [20]), with $H = 3.06 \dots \cdot 10^{10}$.

Whilst the latest and lowest value for H may appear to be a retrograde step, the computations in [21] utilised interval arithmetic and rigorously derived truncation bounds to ensure the results claimed are correct. The two earlier results have the disadvantage that neither has been published in a peer reviewed journal. Furthermore, it is not clear how the computations were set up to avoid problematic accumulation of rounding and truncation errors. This concern was noted in works by Tao [32] (see remarks after Theorem 1.5) and Helfgott [12]. We also mention the result claimed by Franke et al. [9] with $H = 10^{11}$: again, few computational details are given.

We now state our main theorem, which surpasses all aforementioned results.

Theorem 1. *The Riemann hypothesis is true up to height 3 000 175 332 800. That is, the lowest 12 363 153 437 138 non-trivial zeroes ρ have $\Re \rho = 1/2$.*

We note that this independently verifies the results of Gourdon, Wedeniowski and Franke et al. and indeed goes 22% higher than the largest of these.

We have endeavoured to make this paper as short as possible. In §2 we outline some of the computational aspects underpinning Theorem 1. In §3 we mention some results that are improved instantly with Theorem 1.

2 Theory and computation

We used the algorithm described in [21]. In common with all modern partial verifications of the Riemann hypothesis, the algorithm computes values of the completed zeta function on the half line and counts sign changes therein. Each sign change represents a zero of zeta on the half line. Using a variation of Turing’s method (see [34, 35]), we can confirm that all the expected zeroes have been accounted for, so none lie off the half line and the Riemann hypothesis holds in the given range.

We rewrote the original code to utilise Arb [13] in place of MPFI [26] for two reasons. First, Arb is being actively maintained whereas MPFI is not. Second, Arb uses ball arithmetic in place of full interval arithmetic whence there is a space saving of roughly 50%, which make applications more cache friendly.

The other main change to the code that was used to reach $3 \cdot 10^{10}$ in [21] was that we made no attempt to isolate zeroes to any more precision than was absolutely necessary. A key motivation of [21] was to generate a database of rigorously isolated zeroes to high precision, but to do so here would have added to the run time and, in any case, we had nowhere to store that many zeroes. Rather, once we had found a sign change in the completed zeta function indicating the presence of a zero of zeta on the half line, we did not use the machinery of Shannon–Whittaker–Nyquist to “zoom in” on that zero, we merely counted it and moved

were the lowest lying zeroes. A later version of the same slides from 12th November 2003 mentions 561 billion zeroes or $H = 1.53 \dots \cdot 10^{11}$. To add to the confusion, the link [37] gives 900 billion zeroes and this leads to the quoted $H = 2.41 \dots \cdot 10^{11}$.

on. In fact, the default lattice sampling rate we used (about 0.01) was sufficient to isolate 999 997.5 out of every 1 000 000 zeroes.

All computations were performed on the University of Bristol’s Bluecrystal Phase III [1] and the National Computing Infrastructure’s Raijin and Gadi [19] clusters. Both Raijin and Gadi nodes support Hyper-Threading meaning there are two logical processors per core. Since both of these logical devices share the same physical execution resources, one would not expect to see a two times speed up: we found we benefitted to the tune of about 15%. Bluecrystal does not have Hyper-Threading enabled. We also incorporated a minor improvement to our bound for $|\Gamma((\sigma + it)/2)| \exp(\pi t/4)$ (A.2 in [21]) and spent some time optimising the computational parameters to work better 100 times higher up the half line.

In total we used some 7.5 million core hours on 3.6GHz Intel[®] Xeon[®] processors, so each GHz-hour processed a piece of the half line of length about 110 000. For comparison, Wedeniwski [36] reports that the isolation of 561 billion zeros took the equivalent of 2 304 years on 2GHz Pentium[®] 4, so about 3 800 of the half line per GHz-hour whereas Gourdon’s computation took the equivalent of 525 days on a single 2.4GHz Pentium[®] 4, so about 80 000 000 of the half line per GHz-hour. The difference between the Wedeniwski computation and Gourdon’s and our’s shows the power of FFT based algorithms up against vanilla Riemann-Siegel. The fact that Gourdon’s computation was 725 times quicker than ours is down to the higher sampling rate we used (25 per zero in place of 1.2) and the cost of multi precision rigorous numerics compared to hardware floating point.

3 Some instant wins

Where researchers have used Gourdon’s or Platt’s H as their starting point, this independent verification adds weight to their results and gives some explicit improvements. In this section we include some results that are improved either instantly, or at least fairly easily, in light of Theorem 1. We have not endeavoured to furnish an exhaustive list.

3.1 Bounds on primes

It is useful to have explicit estimates on the error term in the prime number theorem. Define $\psi(x) = \sum_{p^m \leq x} \Lambda(n)$, where $\Lambda(n)$ is the von Mangoldt function, and $\theta(x) = \sum_{p \leq x} \log p$. Rosser and Schoenfeld [28] instigated a program of research to bound $\psi(x) - x$ explicitly. The current best results are due to Dusart [7] and Faber and Kadiri [8] for small values of x , Broadbent et al. [3] for intermediate values, and the authors [23] for larger values. These results were based on Gourdon’s value of H which is confirmed by and can be improved improved slightly with our Theorem 1.

Related to these bounds are Bertrand-type estimates: exhibiting a prime in intervals of the form $(x, x + cx]$ for some c and for all $x \geq x_0(c)$. The best results of this type are by Kadiri and Lumley [15], and can now be improved with Theorem 1.

Of course, on the Riemann hypothesis much more is known: Schoenfeld [29] proved that

$$|\psi(x) - x| \leq \frac{1}{8\pi} x^{1/2} \log^2 x, \quad (x > 59). \quad (1)$$

Büthe [5, §7] showed that if the Riemann hypothesis holds up to height H then (1) holds for those x such that $4.92\sqrt{x}/\log x \leq H$. Given this we may note a quick corollary.

Corollary 1. *The following bounds hold in the range indicated*

$$\begin{aligned} |\psi(x) - x| &\leq \frac{\sqrt{x}}{8\pi} \log^2 x, \quad (59 < x \leq 2.169 \cdot 10^{25}), \\ |\theta - x| &\leq \frac{\sqrt{x}}{8\pi} \log^2 x, \quad (599 < x \leq 2.169 \cdot 10^{25}), \\ |\pi(x) - \text{li}(x)| &\leq \frac{\sqrt{x}}{8\pi} \log^2 x, \quad (2657 < x \leq 2.169 \cdot 10^{25}). \end{aligned}$$

We also mention that some results from the expansive article by Bennett et al. [2] can be improved, as can bounds on weighted sums of $\Lambda(n)$ as given by Ramaré [25].

3.2 Zero-free regions and zero density estimates

It is known that there are no zeroes $\rho = \beta + i\gamma$ in the region $\beta \geq 1 - \frac{1}{R \log \gamma}$ for all $\gamma > 3$. The current best value is $R = 5.573412$ by Mossinghoff and Trudgian [18]. This can be improved slightly with Theorem 1 and will be tackled in a future paper of Mossinghoff and the second author, along with the explicit version of the Vinogradov–Korobov zero-free region, which was proved by Ford [10].

Not only is the zeta-function non-zero in regions close to $\sigma = 1$, its reciprocal does not grow too quickly. This has been quantified, explicitly, by the second author in [33]. These results depend not just on the value of H , and hence are improvable by Theorem 1, but also on the size of the zero-free constant R mentioned above.

Let $N(\sigma, T)$ count the number of zeroes with $\beta > \sigma$ and $0 < \gamma \leq T$. Explicit estimates on $N(\sigma, T)$ have been given by Kadiri [14], Kadiri, Lumley, and Ng [16], and Simonič [30]. All of these results could be improved with Theorem 1.

3.3 Oscillations in the prime number theorem

The prime number theorem gives $\pi(x) \sim \text{li}(x)$ and $\theta(x) \sim x$. Littlewood [17] showed that there are infinitely many sign changes in the differences $\pi(x) - \text{li}(x)$ and $\theta(x) - x$. The history of estimating the first sign change of the first of these differences is rich: see [6, 31] for further details. These results, as well as the results of the authors [22] for the first sign change of $\theta(x) - x$ could potentially be improved with Theorem 1. Such an improvement would only be meagre, though, owing to the known small gap between unconditional results and those contingent on the Riemann hypothesis.

The aforementioned results show that the first sign changes cannot be too large. Numerical work shows that the first sign changes cannot be too small, either. Büthe in [4] shows that $\theta(x) < x$ and $\pi(x) < \text{li}(x)$ for $2 \leq x \leq 10^{19}$. We note that these results relied on the value of $H = 10^{11}$ in Franke et al. [9] and so, could also be improved with our Theorem 1.

3.4 The de Bruijn–Newman constant

For $t \in \mathbb{R}$, let

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du, \quad \Phi(u) = \sum_{n=1}^\infty (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) \exp(-\pi n^2 e^{4u}).$$

The de Bruijn–Newman constant Λ is the real number for which all the zeroes of H_t are real when $t \geq \Lambda$. The Riemann hypothesis is equivalent to the statement that $\Lambda \leq 0$. Rodgers and Tao [27] proved that $\Lambda \geq 0$. A history of bounds on Λ is given in [27] and [24].

The 15th Polymath Project [24] contains some calculations with the de Bruijn–Newman constant: the authors prove that $\Lambda \leq 0.22$. We note that we can make an instant, but very mild, improvement on this. The second row in Table 1 on page 65 of [24] shows² that one may take $\Lambda \leq 0.2$ provided one has shown $H > 2.51 \cdot 10^{12}$. This leads to the following.

Corollary 2. *We have $\Lambda \leq 0.2$.*

The next entry in Table 1 of [24] is conditional on taking H a little higher than 10^{13} , which of course, is not achieved by Theorem 1. This would enable one to prove $\Lambda < 0.19$. Given that our value of H falls between the entries in this table, it is possible that some extra decimals could be wrought out of the calculation. We have not pursued this.

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²Note that the X in Table 1 corresponds to $2H$.

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